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The Competitive Effects of Entry: Evidence from Supercenter Expansion†

By Peter Arcidiacono, Paul B. Ellickson, Carl F. Mela, and John D. Singleton*

Coupling weekly grocery transactions with the exact location and opening date of Walmarts over an 11-year period, we examine how Supercenter entry affects prices and revenues at incumbent supermarkets. We find that entry within 1 mile of an incumbent causes a sharp 16 percent drop in revenue, a competitive effect that decays quickly with distance. Surprisingly, despite large cross-sectional differences in supermarket prices by exposure to Walmart, our findings also indicate that Supercenter entry has no causal effect on incumbent prices. This result is robust across many dimensions, including a lack of price response for individual products, and across brands within a category. (*JEL* D22, L81, R32)

Economic frameworks differ in their predictions regarding how retail prices, which play an important role in allocating resources and affecting social welfare, respond to changes in market conditions. In standard models used by economists and government regulators, firms are assumed to set prices to equalize marginal revenue and marginal cost where positive markups are supported via product differentiation (Ackerberg et al. 2007). In contrast, macroeconomists emphasize the role of cost-side frictions in setting prices, such as menu and monitoring costs, or long-term contracts that constrain the rate and frequency with which firms adjust prices. The recent widespread availability of high-frequency supermarket scanner data has yielded a rich empirical literature evaluating the source and scale of these adjustment costs and exploring their role in determining the effects of monetary policy (Klenow and Malin 2010; Eichenbaum, Jaimovich, and Rebelo 2011).[1]

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[1] There is also a tradition, extending at least as far back as the influential case studies of Cyert and March (1963), that views price-setting by firms as "boundedly rational" (Simon 1955, 1962).

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While there exists work examining the degree to which firms react and pass costs through to consumers (e.g., Peltzman 2000; Besanko, Dubé, and Gupta 2005), much less is known empirically about how, and how quickly, firms react to changes in market structure or other "demand shocks" that impact marginal revenue. These kinds of changes are important for understanding the mechanism of price adjustment because, in contrast with changes in costs, they force firms to contend with potentially complex changes in their residual demand curves. From an empirical standpoint, however, identifying the causal impact of demand shocks is challenging since it requires a large number of sizable shocks to reliably infer a consequential impact on pricing, but any such changes are likely to be endogenous, threatening causal inference.

Our empirical strategy thus involves isolating and exploiting a particular set of shocks that are large in magnitude and for which the precise *timing* can be viewed as exogenous: entry by Walmart Supercenter outlets.[2] Walmart's "everyday low price" strategy and 15–25 percent price advantage present a sizable competitive advantage that is consistent across treatments (Hausman and Leibtag 2007; Ellickson, Misra, and Nair 2012). Moreover, retail grocery presents a nearly ideal setting for quantifying price response by incumbents for at least three reasons. First, supermarkets generally set prices above marginal costs, ensuring a wide latitude for competitive price response.[3] Second, supermarket firms were among the earliest adopters of computerized information technology (e.g., scanning registers, electronic data interchange), suggesting that they are as well positioned as any to implement sophisticated pricing strategies. Finally, grocery markets are both spatially localized and differentiated in product space, yielding many "local experiments" that impact both stores and products at different points in the quality spectrum. This bolsters statistical power and provides the variation necessary to rule out a variety of confounding explanations for our baseline results.

To measure the causal impact of Walmart's entry on incumbent pricing, we combine data on the exact geographic locations and opening dates for the universe of Walmart Supercenters with high-frequency transactions data for a representative sample of US supermarkets. Importantly, the high resolution of these unique data allow us to make two contributions to inferring causality. First, we are able to use the confidential addresses of incumbents to examine how the effects depend on the driving distance between the existing supermarket and the entering Supercenter.

[2] Gagnon and López-Salido (2015) pursue a similar approach by examining the price response of US supermarkets to natural disasters, severe weather, and work stoppages due to labor disputes and find small price effects. While these events certainly constitute significant, very likely exogenous, shocks, there are many institutional barriers in place that limit the ability of the firms to change prices in response to such events (e.g., laws and social norms that discourage price gouging, emphasis on "fair pricing," Rotemberg 2011). Moreover, because they also consider only market-wide price responses, they do not capture the highly localized nature of retail competition. This is especially salient in grocery markets since consumers purchase groceries at least once a week (due to perishability) and have historically been reluctant to travel large distances or purchase online.

[3] The ability to price above marginal costs requires a degree of local market power. In the supermarket industry, this pricing power is derived from strategic differentiation and is needed to cover the sizable fixed costs associated with operating a dense network of physical outlets. According to US Census's Annual Retail Trade Survey, gross margins for US grocery stores averaged 28.9 percent in 2006 (midway through our data window), magnitudes that are also in line with those recovered directly from detailed store-level price and cost information (Montgomery 1997, Stroebel and Vavra 2019). We provide additional discussion of the evidence for and sources of local market power in the supermarket industry in Section I.

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Knowing the exact locations is important given the localized nature of retailing (Smith 2006, Figurelli 2013). Second, because we observe weekly transactions at the store level, we are able to exploit the exact timing of Supercenter openings. We observe incumbent revenue and prices immediately prior and just after a Supercenter opening, allowing us to pin down Walmart's causal impact by controlling for its endogenous choice of location.

To establish the magnitude and importance of these shocks, we first examine the revenue impacts of Walmart Supercenter exposure. We find declines in revenue on the order of 16 percent of sales, on average, at the stores most geographically proximate. Supercenter openings are large and disruptive events. In addition, reflecting the local nature of retail competition, we find that revenue effects do not statistically differ from zero beyond a seven-mile exposure distance. At the same time, supporting our research design, we also do not find any evidence of pretreatment trends in revenue but rather revenue declines that immediately follow the opening of a nearby Supercenter. The size and immediacy of the revenue impacts thus suggest a relevant setting to observe competitive price reactions.

Surprisingly, we find that prices at incumbent supermarkets do not react at all to Supercenter entry—either immediately, preemptively, or over the longer term. This result thus stands in sharp contrast to the prior literature examining Walmart entry, which finds considerable price reductions (Basker and Noel 2009, Hausman and Leibtag 2007).[4] It also is surprising given that Supercenter entry affects entry and exit decisions (Arcidiacono et al. 2016).[5] We show that this difference is explained by our empirical strategy of leveraging the timing of Supercenter openings, as opposed to cross-sectional variation in prices, via the inclusion of store fixed effects. Absent store fixed effects—and therefore not accounting for the selective nature of Walmart's entry decisions within a market—we estimate price reactions on the order of 2.5 percent to 3.5 percent, according closely with earlier findings.

Focusing first on possible confounds that threaten causal inference, we examine whether changes in the composition of *products* sold could be masking a measurable price response. For example, firms could adjust pack sizes or product assortments instead. To address this possibility, we extend our empirical approach to UPC-level data, allowing us to examine prices for the same product in the same store before and after Supercenter exposure, and find no evidence for price reactions. We then consider whether changes in the composition of *consumer types* could produce pressures on the direction of price response (e.g., the incentive to raise price to exploit the remaining captive consumers balances exactly the incentive to expand market share by lowering price). This tension, formalized in Chen and Riordan (2008), has been highlighted in several empirical settings.[6] To evaluate the possibility that these

[4] Though it is not the focus of either paper, Matsa (2011) and Ailawadi et al. (2010) also find limited price response to Walmart. In these studies, the null result is due to low precision of the point estimates, whereas the null result we find here is sharp. Furthermore, neither paper addresses the issue of selection in Walmart's choice of location or the role of distance in determining its impact. These aspects play a central role in our analysis.

[5] Our dataset is a sample of the retailer landscape with few exits and entries, so we do not examine this margin here. Ellickson and Grieco (2013) examine incumbent exit using an annual census of retailers.

[6] For example, in the context of air travel, incumbent air carriers reduce ticket prices in response to rival entry (Reiss and Spiller 1989; Borenstein 1989, 1992). The response is particularly pronounced for entry by low-cost carriers like Southwest Airlines (Morrison 2001) and may even take place in anticipation of entry (Goolsbee

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incentives offset, on average, we examine competitive impacts across price tiers and find no price effects despite vertical differentiation of brands within category (and large revenue drops at the bottom of the quality spectrum consistent with Walmart's low-tier, low-price positioning).

Turning to the interpretation of our results, some potential explanations for the lack of price response are effectively ruled out by the setting. Like DellaVigna and Gentzkow (2019), we believe that menu costs are not a factor, as supermarkets frequently adjust prices to implement sales. Monitoring costs are also unlikely to play a role, since Supercenter entry nearby and its corresponding effect on revenue would be impossible not to notice. Similarly, the well-documented existence of significant price-cost markups in grocery retail, even relative to Walmart's large price advantage, implies that firms are unlikely to be precluded from cutting prices by a hard cost constraint. Upward adjustments of prices, which as we discuss in Section I may be optimal should the incumbent become a monopolist on the residual demand curve, are further unlikely to be constrained.[7] We find it more plausible that firms could be constrained by long-term or otherwise restrictive contracts with manufacturers (Villas-Boas 2007), who have less incentive to respond to local retailer entry. To address this possibility, we therefore examine price response separately for branded products and for store brands for which prices are entirely under the retailer's control. We do not find evidence of price reactions for either product type. Another plausible explanation concerns the multimarket nature of chain retailing. In particular, chains with many widely dispersed outlets might set prices regionally or even centrally to economize on advertising expenses or to leverage other scale or scope economies (Hitsch, Hortacsu, and Lin 2017; DellaVigna and Gentzkow 2019; Adams and Williams 2019), thereby blunting the response to competitive entry impacting local stores. However, our data reveal that even local chains (e.g., those with only a handful of stores) do not respond to Supercenter entry by adjusting prices, casting doubt on the empirical relevance of these factors as well.

We argue that the explanation most consistent with our findings is that supermarket firms routinely employ "rule of thumb" cost-plus pricing policies, a form of managerial inattention. Since entry by Walmart does not impact costs, it does not trigger a price reaction from incumbents. We discuss the evidence for this and for alternative remaining interpretations before concluding the paper. While our data, lacking information on costs, do not allow us to test this hypothesis directly, this interpretation is consistent with other recent empirical studies of supermarket pricing. In particular, Eichenbaum, Jaimovich, and Rebelo (2011) exploit detailed information on both wholesale costs and retail markups for a very large supermarket

and Syverson 2008). This is consistent with a very strong competitive price effect, likely reflecting the relative homogeneity of the underlying consumer product. On the other hand, upon entry by generic drugs, incumbent manufacturers of branded pharmaceuticals typically *raise* prices (Frank and Salkever 1992). This is despite the fact that generic entrants are required to establish bioequivalence by the FDA (meaning that "true" product differentiation is minimal). This behavior may be an optimal response to the shift in the distribution of consumer types toward those with higher willingness to pay for the branded products (Frank and Salkever 1997), a mechanism that likely operates through access to insurance.

[7] Notably, using a similar empirical approach and dataset to Basker and Noel (2009), Courtemanche and Carden (2014) examine the impact of entry by club stores (Costco and Sam's Club) on grocery prices. They find that Costco entry is associated with higher prices at incumbent grocers, while entry by Sam's Club has no effect.

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chain to show that grocery prices rarely change unless triggered by an accompanying change in costs. Prices are set with a relatively long duration, aimed at keeping markups within a fairly narrow target range. McShane et al. (2016) provide similar evidence using data from another retailer.[8]

Our findings contribute to several strands of literature. First, we contribute to prior work that studies the economics of supercenters and examines the implications of the competitive advantages enjoyed by Walmart.[9] Our analysis reveals that Walmart favors locating new Supercenters near relatively high-revenue and low-priced competitors (a selection rule that we show biases prior estimates of competitive effects of Walmart). Our findings also have important implications for the competitive welfare gains from Walmart Supercenter expansion. Hausman and Leibtag (2007) argue that fully one-fifth of the consumer benefits of Supercenter expansion is due to price reductions at rival stores. In contrast, our results indicate no such competitive price response, implying that the actual welfare increase is substantially smaller. We also find no evidence of an assortment or repositioning response. While we cannot rule out responses along all dimensions of quality—Matsa (2011) presents evidence that stockouts decrease at rival firms when Walmart enters—our null results suggest that the primary impact of entry is not lower incumbent prices (or different assortments) but simply lower sales volume at incumbent stores. This is likely to lead to an increased likelihood of store failure, consistent with Ellickson and Grieco (2013), suggesting that the competitive reaction may primarily occur along the extensive rather than intensive margin.

Our results also contribute to a small but growing literature on behavioral aspects of firm decision-making and their implications for market efficiency. For example, both Hitsch, Hortacsu, and Lin (2017) and DellaVigna and Gentzkow (2019) document that price variation within retail chains across markets is far more muted than would be predicted by local market conditions, while Blake, Nosko, and Tadelis (2015) find vast overspending by eBay on search advertising. Shapiro (forthcoming) also concludes that pharmaceutical firms over-advertise.[10] Lastly, our results contribute to recent empirical work in macroeconomics focused on understanding the sources of nominal rigidities and the stickiness of price adjustment (Blinder et al. 1998, Bils and Klenow 2004, Klenow and Malin 2010). Our contribution lies in highlighting the relative importance of demand versus cost shocks in determining price adjustment in retail settings.

[8] There is also a large volume of both anecdotal and survey-based evidence documenting the widespread use of cost-plus pricing in both industrial and retail settings (Blinder et al. 1998; Noble and Gruca 1999; Phillips 2005; Watson, Wood, and Fernie 2015).

[9] See, e.g., Basker (2007), Holmes (2011), Jindal et al. (2018), and papers examining Walmart's impact on a variety of market outcomes, including wages and employment (Basker 2005; Neumark, Zhang, and Ciccarella 2008), market structure (Foster, Haltiwanger, and Krizan 2006; Jia 2008; Zhu, Singh, and Manuszak 2009; Ellickson, Houghton, and Timmins 2013), rival responses (Ailawadi et al. 2010, Matsa 2011), and consumer welfare (Hausman and Leibtag 2007; Atkin, Faber, and Gonzalez-Navarro 2018).

[10] Papers that examine the role of managerial ability in driving departures from optimizing behavior also include Bloom and Van Reenen (2007); Hortaçsu and Puller (2008); Goldfarb and Xiao (2011); and Ellison, Snyder, and Zhang (2018).

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I. Institutional Background and Data

US supermarkets sold just over $700 billion in grocery products across roughly 38,000 individual stores in 2018. A relatively mature industry that remains predominantly brick-and-mortar, the grocery trade is served by a heterogeneous collection of chains and sole proprietorships that vary in size from a single store to a couple thousand outlets. The median store includes roughly 41,000 feet of floor space and stocks about 30,000 products.[11] The majority of sales are accounted for by large, vertically integrated chains that sell a mix of branded products (e.g., Coca-Cola) and store-brand private labels (e.g., retailer-branded soda). Most branded products are sourced directly from their manufacturers and then distributed via a given retail chain's hub-and-spoke distribution network. Store brand products are cheaper to source because the upstream market is highly competitive (there is no single brand to negotiate with) and offer the retailer greater autonomy regarding final pricing decisions.[12]

Supermarket firms and their outlets are differentiated both horizontally and vertically.[13] The most obvious horizontal element is location, though assortments are sometimes tailored to particular demographic groups. For example, stores are more likely to stock larger package sizes in markets where consumers have greater storage capacity (e.g., bigger houses with larger pantries). The key vertical aspects of differentiation include overall store size, product assortment, and quality/service levels. Targeting relatively distinct segments of the consumer population partially insulates rival firms from direct price competition, conferring a degree of local market power that is necessary to cover high levels of fixed operating costs. Consequently, most grocery chains are relatively specialized, selecting product assortments and service levels that appeal to different consumer types. This mainly operates through income (e.g., Whole Foods versus Aldi). One implication of retailer differentiation is that chains' entry decisions (site locations) clearly exhibit selection as evidenced in our empirical analysis below.

From a competition standpoint, differentiation is the key mechanism that allows supermarkets to set prices sufficiently far above marginal cost to cover their fixed outlays. Absent the exercise of local monopoly power, rival firms would be unable to cover the large fixed costs associated with operating distribution networks as well as staffing and maintaining their physical outlets. While often characterized as a "low margin" industry, the grocery trade (like most modern retail sectors) is actually a high *gross* margin, low *net* margin business, a distinction that is the source of some confusion. Net margins, which capture overall profitability (and are used for financial valuation), are on the order of 1–3 percent, reflecting strong competitive pressure at the entry stage.[14] Gross margins, which are the relevant

[11] Industry figures are extracted from Food Marketplace Inc.'s "Supermarket Facts" web portal: https://www.fmi.org/our-research/supermarket-facts.

[12] We leverage this distinction in our analysis of potential explanations for our findings.

[13] Products are horizontally differentiated when they feature characteristics for which differing consumers have distinct tastes (e.g., a location near your home), while products are vertically differentiated when all consumers agree on their relative ranking but perhaps not in their willingness to pay for higher quality (e.g., organic produce).

[14] Montgomery (1997) reports net margins of 3 percent in 1993, while recent industry trade publications put the number closer to 1.5 percent (https://www.fmi.org/our-research/supermarket-facts/grocery-store-chains-net-profit).

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object for assessing pricing constraints following competitive entry (i.e., the scope for adjusting prices downward), are much higher. For example, Stroebel and Vavra (2019, 1434) note that they "directly observe an average markup of roughly 45 percent for our large anonymous retailer," a large US grocery chain for which detailed cost as well as price information is available from 2004 to 2007. Gross margins reported in other studies using different datasets are broadly similar.[15]

In practice, supermarket retailers differ in their ability to target distinct consumer segments, meaning that they serve heterogeneous populations that at least partly overlap with those of their rivals. Thus, when Walmart enters a new market, incumbent grocers will differ in their degree of exposure to the Supercenter's target population, leading to a shift in the composition of consumer types that make up their residual demand curve. Profit maximization implies managers should adjust prices to equate marginal costs to the new levels of marginal revenue.[16] However, due to differentiation, there are two potentially offsetting forces that determine whether the new marginal revenue curve is steeper or flatter than the prior one (Chen and Riordan 2008). First, there is a "market share" effect: firms face pressure to compete for shared consumer types by lowering prices. Second, there is a "price sensitivity" effect: consumers who remain loyal to the incumbent firm are presumably less price sensitive. This instead creates an incentive to raise prices, all else held equal. Importantly, these offsetting forces operate heterogeneously even within a given outlet because different products are targeted to different consumer segments, a feature that we exploit below in our analysis.

A. *Data Sources and Summaries*

Our analysis combines two primary data sources. The first dataset, provided by the retail consulting firm Trend Results (www.trendresults.com), contains the exact opening date and address of every Walmart in operation through 2011. We restrict our attention to the 3,150 Walmart stores designated as Supercenters that carry a full grocery line.[17] These records are used to link the timing of Supercenter openings with associated pricing and other outcomes at nearby incumbent supermarkets.

Our second data source is the IRI marketing dataset, which records weekly store-level sales at a nationally representative sample (roughly 10 percent) of

[15] The average store-level *markup* over marginal cost $\left((p-c)/c\right)$ reported by Stroebel and Vavra (2019) translates to an average *margin* over price $\left((p-c)/p\right)$ of roughly 31 percent. This accords quite closely with data from the Census's Annual Retail Trade Survey, which reports average gross margins for grocery stores (NAICS 4451) over the same period of 28.5 percent. Examining a single category (orange juice) from the Dominick's Finer Foods dataset for 1989–1994, Montgomery (1997) reports an average product-level gross margin of 25 percent. Moreover, the smallest product-level margin he finds is 19.4 percent, suggesting that even low-end products have substantial leeway to lower price. Nakamura (2008) reports store-level margins of 28.3 percent based on data from the US Bureau of Labor Statistics.

[16] In general, this optimization problem is quite complicated because the relevant first-order condition includes terms that capture the substitution to alternative products in the same store and chain (which the focal firm should internalize) as well as substitution to rivals. Thomassen et al. (2017) estimate a spatial model of this type using data from the UK grocery industry.

[17] While these records provide the universe of Walmart store openings since 1962, they do not identify Walmarts that were expanded into Supercenters or when such an expansion took place; 543 traditional Walmarts (i.e., non-Supercenters) remained at the time our data were gathered (159 of which opened in 2001 or later).

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supermarkets (Bronnenberg, Kruger, and Mela 2008).[18] The data cover the period from 2001 through 2011 (11 years in total). The data include the transaction records for 2,450 incumbent supermarkets during the 573 weeks (a median of 293 weeks per store) that span the sample period. About 60 percent of the IRI supermarkets represent (non-Walmart) national chains, and another 6 percent do not belong to a chain at all, while the remaining stores belong to regional or local chains.[19] Geographically, the supermarkets are spread across 50 IRI markets, which are approximately equivalent to Metropolitan Statistical Areas (MSAs). We augment the transactions records with the physical address of each supermarket. Table A1 in the Appendix presents additional summaries of the incumbent stores, including the demographics of the markets they draw from.[20] Each record in the IRI data contains a Universal Product Code (UPC) identifier, quantity purchased, and the purchase price of the item at a given supermarket in a given week. Transactions data for 15 food product categories are used to construct our dependent variables, which we describe next.[21]

For each IRI supermarket, we construct weekly revenue and price series by product category. Price series are constructed by dividing total weekly revenues across all UPCs sold in the category by the total category volume sold across those UPCs.[22] We use the definitions of volume equivalent units for each category (e.g., 1 unit $=$ 16 oz. of coffee) provided by IRI. Thus, the price of product category c in week t at store s is the total revenue divided by the total sales volume in category c for that week:

$$P_{st}^{c} = \frac{\sum_i P_{ist}^{c} \times Q_{ist}^{c}}{\sum_i Q_{ist}^{c} \times V_{ist}^{c}}. \tag{1}$$

In this equation, i indexes UPC, P_{ist}^{c} represents the actual transaction price of i in store s during week t, V_{ist}^{c} is the volume weight of that UPC, and Q_{ist}^{c} denotes the total unit sales of the UPC. We deflate all prices to January 2001 dollars using the CPI.

Summary statistics for the incumbent supermarkets' revenue and prices are provided in Table 1. In a given week, incumbent supermarkets generate nearly \$45,000 (which annualizes to \$2.3 million) in revenue, on average, across the 15 food categories considered. These product categories represent about 58 percent of all revenue recorded and released by IRI. Notably, there is considerable variation in weekly store-level revenue, with the ninetieth percentile representing

[18] IRI considers several factors, which include the homogeneity of retailers' stores, the cost of acquiring retailer information, the size of the retailer, and whether the retailer is willing to provide data, for inclusion in the sample. Although the sample is not randomly drawn, IRI strives to maximize coverage and market representativeness in its composition. One limitation of IRI's sampling design, noted earlier, is that we are unable to reliably examine effects on entry and exit.

[19] Because Walmart is not included in the IRI data, a limitation is that we cannot separately estimate impacts for the set of products that are carried in Supercenters.

[20] Demographic estimates for 2007 are provided by Applied Geographic Solutions via IRI for a two-mile radius around 2,012 of the supermarkets.

[21] We exclude milk, a product category subject to substantial price regulations, and beer, which is not sold in all supermarkets due to varying regulatory environments.

[22] A similar price index and panel data approach was used by both sides to determine the extent to which Office Depot and Staples constrained each other's pricing in the US Federal Trade Commission's challenge of the Staples/Office Depot merger (Ashenfelter et al. 2006).

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TABLE 1—SUMMARIES OF INCUMBENT SUPERMARKET OUTCOMES

	Units	Observations	Mean	SD	p10	Median	p90
Revenue	Dollars	756,097	44,738	26,949	18,099	38,378	79,275
Carbonated beverage price	$/192 oz.	755,078	3.69	0.56	3.07	3.62	4.40
Coffee price	$/16 oz.	755,076	4.43	1.37	2.88	4.23	6.29
Cereal price	$/16 oz.	755,282	2.64	0.37	2.22	2.61	3.08
Frozen dinners price	$/16 oz.	681,853	2.91	0.52	2.29	2.88	3.59
Frozen pizza price	$/16 oz.	755,002	2.78	0.53	2.17	2.72	3.44
Hotdog price	$/16 oz.	754,935	2.27	0.59	1.56	2.24	3.02
Margarine/butter price	$/16 oz.	755,107	1.36	0.43	0.87	1.28	1.94
Mayonnaise price	$/16 oz.	755,026	1.55	0.32	1.17	1.53	1.97
Mustard/ketchup price	$/16 oz.	755,387	1.15	0.30	0.81	1.10	1.54
Peanut butter price	$/16 oz.	755,383	1.74	0.27	1.43	1.72	2.08
Salty snack price	$/16 oz.	754,635	3.08	0.39	2.61	3.06	3.56
Soup price	$/16 oz.	755,235	1.40	0.22	1.14	1.38	1.69
Spaghetti sauce price	$/16 oz.	755,051	1.12	0.27	0.84	1.08	1.43
Sugar substitute price	$/1 oz.	755,422	0.67	0.16	0.50	0.65	0.87
Yogurt price	$/1 pint	754,724	1.46	0.21	1.21	1.44	1.71

Notes: Table summarizes revenue and price by product category at the incumbent supermarket-week observation level. Revenue is for only the summarized categories. Values are in January 2001 dollars per CPI.

nearly twice the average. There is also substantial price variation. In the median store-week, 16 oz. of coffee costs $4.23 but at the ninetieth percentile costs $6.29—around 50 percent more expensive. The high-frequency variation in these series (week-to-week) allows us to isolate changes at incumbents immediately following a Supercenter opening.

Linking the Supercenter openings with the IRI supermarket data involves two steps. First, we create Supercenter-IRI store matches if the straight line ("crow's flight") distance between any pair is less than 15 miles. Of the 1,055 Supercenters that Walmart opened in the United States during our sample period (between January 2001 and the end of 2011), this decision rule matches 481 unique Supercenters with at least 1 supermarket in the IRI dataset.[23] Second, we compute the driving distance between the Supercenters and the incumbents' locations using Google's mapping software API. Google Maps measures driving distance by calculating the fastest route by car between two locations, providing an accurate measure of the navigable spatial distance between competing stores. Our final merged sample contains 756,097 store-week observations.

Merging these data sources provides both cross-sectional and panel variation in Supercenter exposure. Incumbent supermarkets' exposure to Walmart Supercenters in our sample is summarized in Table 2. The variable #WM represents a count of all Supercenters operating within 11-miles driving distance of any supermarket in the IRI dataset, which we set as the maximum exposure distance.[24] The total count is then binned into driving distance bands centered around each incumbent supermarket. In the average store-week, an incumbent IRI store is exposed to 1.6 Supercenters. Additionally, counts of the exposures during our sample categorized by distance band and entry order (i.e., first ever exposure, second, etc.) are

[23] In total, 1,311 of the 3,150 Supercenters ever opened before 2012 are matched to at least 1 IRI supermarket.
[24] Note that driving distance miles are greater than or equal to "crow's flight" miles by construction.

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Table 2—Summary of Walmart Supercenter Exposures

	Mean	SD	p10	p25	Median	p75	p90
#WM	1.60	1.63	0	0	1	2	4
#WM $<$ 1 mile	0.04	0.20	0	0	0	0	0
1 mile $\leq$ #WM $<$ 3 miles	0.19	0.41	0	0	0	0	1
3 miles $\leq$ #WM $<$ 5 miles	0.26	0.51	0	0	0	0	1
5 miles $\leq$ #WM $<$ 7 miles	0.32	0.60	0	0	0	1	1
7 miles $\leq$ #WM $<$ 9 miles	0.37	0.67	0	0	0	1	1
9 miles $\leq$ #WM $<$ 11 miles	0.42	0.71	0	0	0	1	1

Note: Table summarizes incumbent supermarket exposure to Walmart Supercenters overall and by driving distance; there are 756,097 incumbent store-week observations.

Table 3—Observed Supercenter–Incumbent Supermarket Exposures

Exposure	1st	2nd	3rd	4th	5th	6th	7th	Total
#WM $<$ 1 mile	7	15	6	0	0	1	1	30
1 mile $\leq$ #WM $<$ 3 miles	20	42	25	12	6	5	0	110
3 miles $\leq$ #WM $<$ 5 miles	24	47	35	30	11	10	9	166
5 miles $\leq$ #WM $<$ 7 miles	30	64	54	42	19	18	7	234
7 miles $\leq$ #WM $<$ 9 miles	50	71	70	46	39	16	14	306
9 miles $\leq$ #WM $<$ 11 miles	55	81	79	64	28	21	16	344
Total	186	320	269	194	103	71	47	1,190

Notes: Table summarizes observed Walmart Supercenter entries by driving distance band and incumbents' order of exposure. For example, the first cell of the first column indicates that we observe seven Supercenter openings within one mile that are the incumbent's first exposure at any distance during our sample.

presented in Table 3. In the 1- to 3-mile driving distance band, for example, we observe 110 exposure events in our sample, 20 of which were the incumbents' first ever exposure to a Supercenter at any distance and 42 of which were their second. During the 11-year sample, 1,190 total exposure events are observed within the maximum 11-mile driving distance. Our empirical model leverages this variation in both timing and distance.

Our analysis aims to uncover the causal effects of Supercenter exposure on incumbent outcomes. As a prelude to the full analysis, however, we first examine the average differences in incumbent supermarket outcomes immediately before and after a Supercenter opening. Table 4 therefore reports revenues and prices for exposed incumbents by driving distance band for the eight weeks immediately prior to an opening (columns labeled "Pre") and the eight weeks following (columns labeled "Post"). The summaries are restricted to the incumbents' nearest Supercenter exposure. The table reveals two main descriptive results. First, as a comparison of the "Pre" columns across the distance bands shows, there is a robust association between lower average prices and closer treatment exposures. In other words, supermarkets located nearer to a Walmart Supercenter tend to be lower-priced even in the weeks immediately leading up to a Supercenter opening. For instance, 16 oz. of coffee costs \$3.71 at stores exposed within 1 mile and \$4.58 (20 percent higher) at stores exposed between 5- to 7-miles driving distance. Importantly, this association may

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Table 4—Incumbent Outcomes for 8 Weeks Pre– and Post–Nearest Supercenter Exposure

	Less than 1 mile		1 to 3 miles		3 to 5 miles		5 to 7 miles	
	Pre	Post	Pre	Post	Pre	Post	Pre	Post
Revenue	55,193	46,494†	45,305‡	42,971	40,800‡	39,301	47,166‡	45,980
Prices:								
Carbonated beverage	3.66	3.71	3.66	3.67	3.61	3.63	3.69	3.70
Coffee	3.71	3.83	4.03‡	4.13	4.39‡	4.41	4.58‡	4.47
Cereal	2.50	2.48	2.57‡	2.56	2.63‡	2.64	2.70‡	2.71
Frozen dinners	2.79	2.82	2.90‡	2.90	2.90‡	2.90	2.96‡	2.95
Frozen pizza	2.64	2.58	2.68	2.67	2.80‡	2.81	2.90‡	2.89
Hotdogs	2.05	2.05	2.21‡	2.19	2.29‡	2.28	2.38‡	2.32
Margarine/butter	1.14	1.13	1.22‡	1.23	1.35‡	1.36	1.42‡	1.40
Mayonnaise	1.46	1.46	1.46	1.48	1.56‡	1.57	1.60‡	1.59
Mustard/ketchup	1.05	1.03	1.10‡	1.10	1.15‡	1.16	1.20‡	1.18
Peanut butter	1.64	1.61	1.68‡	1.70	1.77‡	1.76	1.80‡	1.81
Salty snacks	2.89	2.90	2.97‡	2.99	3.02‡	3.05	3.13‡	3.12
Soup	1.34	1.34	1.38‡	1.36	1.46‡	1.43†	1.46‡	1.42†
Spaghetti sauce	1.05	1.02	1.08‡	1.08	1.13‡	1.12	1.17‡	1.15
Sugar substitutes	0.62	0.62	0.63	0.63	0.69‡	0.68	0.69‡	0.69
Yogurt	1.37	1.37	1.41‡	1.41	1.46‡	1.46	1.51‡	1.50
Observations	235	248	792	793	697	712	444	450

Notes: Table compares revenue and prices at incumbent supermarkets eight weeks pre- and eight weeks post-Supercenter entry by driving distance. Values calculated for each incumbent's nearest exposure. *N* represents revenue store-week observations (first row).

† Indicates Post differs from Pre with $p < 0.05$.

‡ Indicates difference from Pre at less than 1 mile band with $p < 0.05$.

reflect the causal effect of Supercenter exposure or Walmart's selection of where to locate. The second finding in Table 4 is that the average within-store, pre/post differences in prices are small and almost always statistically insignificant. In other words, at least in the immediate eight-week windows presented in the table, it is difficult to discern obvious price effects of Supercenter exposure. As an example, cereal prices are on average $2.50 per 16 oz. unit before a Walmart opening and $2.48 post-opening in the less-than-1-mile driving distance band. This descriptive finding suggests little price response, though this may be confounded by dynamics not captured in the snapshot presented.[25] These within-store (post-/preexposure) differences by driving distance are a key element of our empirical model, described in detail in the next section.

II. Empirical Model

Our empirical model exploits the sample's unique spatial and temporal variation to estimate the causal effects of Supercenter exposure on incumbent outcomes. We apply the model to both logged revenue and logged prices of incumbent supermarkets.

[25] It may be, for example, that incumbents anticipate Walmart's arrival by lowering prices in advance or, conversely, do not adjust prices until later.

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To simplify exposition, both are represented by y_{st}, but it should be understood that the estimands differ across the two outcome measures. Our estimating equation can be written generally as

$$y_{st} = \sum_{j} \beta\big(D_{sj}\big) WM_{jt} + \epsilon_{st}. \tag{2}$$

The variable WM_{jt} is an indicator for the presence of the jth Supercenter at time t. The treatment effect of interest, $\beta\big(D_{sj}\big)$, is written as a function of the driving distance, D_{sj}, between the incumbent supermarket and Supercenter j.[26] We assume that there is a distance $\bar{D}$ at which the effects of Walmart entry are zero. Finally, ϵ_{st} represents all other factors that influence y_{st}.

The identification of $\beta\big(D_{sj}\big)$ is confounded by a potential correlation between ϵ_{st} and WM_{jt}. As Walmart deliberately selects where to locate, it is implausible that this correlation is zero. Taking advantage of the variation in the timing of Supercenter entry across incumbents, we specify ϵ_{st} as a function of store fixed effects, π_s; time dummies, λ_t; a market-specific time trend; and a residual error term, υ_{st}.[27] In particular,

$$\epsilon_{st} = \pi_s + \lambda_t + \gamma_{m(s)}\big(t\big) + \upsilon_{st}, \tag{3}$$

where $m\big(s\big)$ denotes the market for store s. Our identifying assumption is that the timing and location of Walmart entry are uncorrelated with υ_{st} conditional on the store fixed effects, time dummies, and market trends. To formalize this notion, denote by μ_{st} the store fixed effect for s, the time dummy at t, and the market trends. Letting WM_{st} represent the set of Supercenters that incumbent s is exposed to at time t, our assumption is that υ_{st} is conditionally uncorrelated with Walmart entry:

$$E\big[\upsilon_{st} | \mu_{st}\big] = E\big[\upsilon_{st} | \mu_{st}, WM_{st}\big] = 0.$$

We approximate $\beta\big(D_{sj}\big)$ in equation (2) using discrete driving distance bands around incumbent supermarkets' locations. This specification is implemented with a set of indicator variables that take on a value of one when store s is exposed to Supercenter j within distance band b. The use of discrete driving bands lets the data determine the distance, $\bar{D}$, at which the effect of exposure on incumbent outcomes attenuates to zero. In practice, we set an upper bound of 11 miles in driving distance beyond which an incumbent is considered "not exposed" to a particular Supercenter opening. Denoting the B distance bands by D_{sj}^{b} and substituting in for ϵ_{st} using (3), (2) becomes

$$y_{st} = \sum_{b=0}^{B} \beta_b \sum_{j} D_{sj}^{b} WM_{jt} + \pi_s + \lambda_t + \gamma_{m(s)}\big(t\big) + \upsilon_{st}. \tag{4}$$

[26] Note that equation (2) assumes that the effect of a particular Supercenter entry on the outcome of interest does not depend on the number of previous exposures.

[27] We also explore specifications that allow for separate trends for each level of total exposures to Supercenters by the end of our sample. We specify these trends to be quadratic.

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This empirical model can be understood as a generalization of a difference-in-difference approach where contemporaneous changes in outcomes are compared between supermarkets that are treated by a Supercenter and stores that are not. For a given Supercenter opening, the control stores include both stores never exposed and stores exposed earlier or later over the duration of the data. The distance band specification extends this logic to also compare stores treated at different driving distances.

III. Main Results

This section reports estimates from the incumbent supermarket revenue and price models. In addition to estimating equation (4), we compare the results with estimates that rely on cross-sectional variation and examine robustness to different specifications of the trends. To check for pretreatment trends that may confound causal inference, we also estimate "event study" specifications, described below. To help establish the relevance of our research design examining incumbent supermarkets' response to the "demand shock" of Supercenter entry, we examine revenue first before turning to competitive effects on prices.

A. *Revenue Effects*

Using log revenue as the dependent variable in equation (4), Table 5 presents the estimates of the effect of Supercenter exposure on incumbent supermarkets' revenue. The first column uses cross-sectional differences (along with the panel variation) in revenue by Supercenter exposure, reporting a negative association in some driving distance bands but no clear pattern overall. In column 2, we include IRI market and supermarket chain fixed effects to compare differentially exposed incumbents in the same chain and market. These results indicate large revenue effects that decay quickly with the exposure distance. At 1 mile, the effect is slightly under 12 percent.

By incorporating store fixed effects, column 3 then applies the empirical model's difference-in-difference approach of exploiting differential timing of Supercenter exposure. The results indicate that entry causes drops in supermarket revenue of over 16 percent when Walmart locates within 1 mile. As the estimated revenue effects at close distances increase significantly when store fixed effects are included, the difference suggests that Walmart locates next to high-revenue stores. Absent store fixed effects, the number of Walmarts within a one-mile distance band serves as a proxy for high-revenue stores. Revenue effects again attenuate sharply with distance, indicating considerable localization in the exposure impact. Within 1- to 3-miles driving distance, exposure depresses revenue by 7 percent, less than half of the effect at under 1 mile. At 3 to 5 miles, the effect falls to 5 percent. Beyond 5 miles, Walmart's effect on incumbent revenue cannot be rejected as different from 0.[28] The final two columns add IRI market-specific and treatment group-specific trends in revenue respectively. The treatment group trends allow

[28] Note this decay is well short of the 11-mile upper bound we set for exposures.

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Table 5—Competitive Revenue Effect of Supercenter Openings

	(1)	(2)	(3)	(4)	(5)
#WM < 1 mile	−0.00966	−0.115	−0.162	−0.160	−0.154
	(0.0697)	(0.0367)	(0.0328)	(0.0341)	(0.0338)
1 mile ≤ #WM < 3 miles	−0.0688	−0.0615	−0.0674	−0.0769	−0.0716
	(0.0300)	(0.0178)	(0.0259)	(0.0248)	(0.0255)
3 miles ≤ #WM < 5 miles	−0.0190	−0.00899	−0.0497	−0.0505	−0.0417
	(0.0222)	(0.0141)	(0.0161)	(0.0158)	(0.0146)
5 miles ≤ #WM < 7 miles	−0.0522	−0.0183	−0.0188	−0.0269	−0.0253
	(0.0198)	(0.0121)	(0.0141)	(0.0144)	(0.0150)
7 miles ≤ #WM < 9 miles	−0.0263	−0.0219	−0.0153	−0.0220	−0.0176
	(0.0173)	(0.0114)	(0.0140)	(0.0133)	(0.0127)
9 miles ≤ #WM < 11 miles	−0.00855	−0.0110	−0.0188	−0.0174	−0.0149
	(0.0169)	(0.0113)	(0.0122)	(0.0116)	(0.0112)
Period fixed effects	Yes	Yes	Yes	Yes	Yes
Market and chain fixed effects	No	Yes	No	No	No
Treat group fixed effects	No	No	No	No	No
Store fixed effects	No	No	Yes	Yes	Yes
Market trends	No	No	No	Yes	Yes
Treat group trends	No	No	No	No	Yes

Notes: Table presents results of regressing incumbent supermarket log revenue on Walmart Supercenter exposure. There are 756,097 store-week observations. Standard errors clustered by store in parentheses. Treatment groups are defined by total Supercenter exposures. Markets are IRI designated markets. All trends are quadratic.

revenues to trend differently based on the total number of Supercenter exposures an incumbent experiences.[29] Adding these controls has little effect on the coefficients on interest, suggesting that store fixed effects are the most material controls.

Causal inference for the estimates is justified under the assumption of conditionally exogenous entry timing. While the validity of this assumption cannot be tested directly, trends in exposed stores' revenue prior to a Supercenter exposure are informative in this regard. If the treated stores and control stores have similar trends leading up to exposure, we fail to reject the "common trends" assumption. To implement this test, we estimate an "event study" specification by modifying our estimating equation to interact the treatment effect of a Supercenter with the time until or since its opening week:

$$(5)\quad y_{st} = \sum_{k=-13}^{13} \beta_k \sum_j \mathbf{1}\{W_{jt} = k\}\mathbf{1}\{D_{sj} \leq 7\} + \sum_{b=1}^{B} \beta_b \sum_j D_{sj}^b WM_{jt} + \mu_{st} + \upsilon_{st},$$

where W_{jt} indexes the number of eight-week windows period t is from Supercenter j's opening date. We restrict the pretrends and posttrends to exposures within 7 miles and bin weeks around the entry date into windows up to 2 years (13 windows)

[29] In other words, unexposed stores have their own trend, one-time exposed stores have their own trend, and so forth.

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pre- and postopening.[30] We choose 7 miles based on the prior results showing the revenue effects are not statistically different from 0 at this distance and beyond. We control for the driving distance of the exposure by continuing to include the exposure counts for all driving distance bands except less than 1 mile ($b = 0$) as regressors. The β_k estimates therefore can be interpreted as the revenue effect in a given window k for an exposure within 1 mile. The specification thus allows the magnitude of Walmart's effect postentry to vary by distance but pools exposures of different distances to estimate the trend components and test for pretrends. Note that this estimation also uncovers how the magnitude of the treatment effect varies postentry.

Online Appendix Table A1 presents the detailed results, finding neither trends leading up to opening nor appreciable changes in the magnitude of the treatment effect postexposure. This is depicted visually in panel A of Figure 1, which plots the β_k point estimates and confidence intervals from estimating (5). Prior to entry (indicated by 0 on the horizontal axis), differences in revenue relative to more than two years prior to entry are not statistically different from zero. Thus, the estimates indicate that the revenue of treated and control stores does not trend differentially prior to treatment. After a Supercenter opening, however, treated incumbent supermarkets (within 1-mile driving distance) experience around 15 percent revenue drop on average that persists over time, with some small attenuation after a year and a half. Panel B of Figure 1 visually displays estimates corresponding to a more flexible specification that allows the trends components to be distance band-specific. The results likewise suggest no pretrends and indicate sharp revenue impacts upon Supercenter entry that are monotonic in treatment distance. In sum, the figures highlight the significant causal, though spatially bounded, consequences of Supercenter exposure on the revenue of competitor supermarkets. An opening of a Supercenter thereby constitutes a significant shock to the local competitive environment. We now examine how these rivals respond in prices.

B. *Price Effects*

To examine the causal effect of Supercenter exposure on incumbent supermarkets' prices, we modify our estimating equation to estimate the effects across all categories by pooling the individual logged price series:

$$y_{st}^{c} = \sum_{b=0}^{B} \beta_b \sum_{j} D_{sj}^{b} WM_{jt} + \pi_s^c + \lambda_t^c + \gamma_{m(s)}^{c}(t) + \upsilon_{st}^{c}. \tag{6}$$

Superscript c indexes the food product categories. Pooling the individual price series obviates the need to construct a store price index from the individual category series.

The results of estimating (6) are presented in Table 6. The baseline estimates in column 1, which leverage cross-sectional differences in prices with exposure distances, indicate large and statistically significant price effects. For example, an

[30] We include a "window" for 105 weeks *or more* postentry and treat windows for prior to 104 weeks as the excluded category in the regression.

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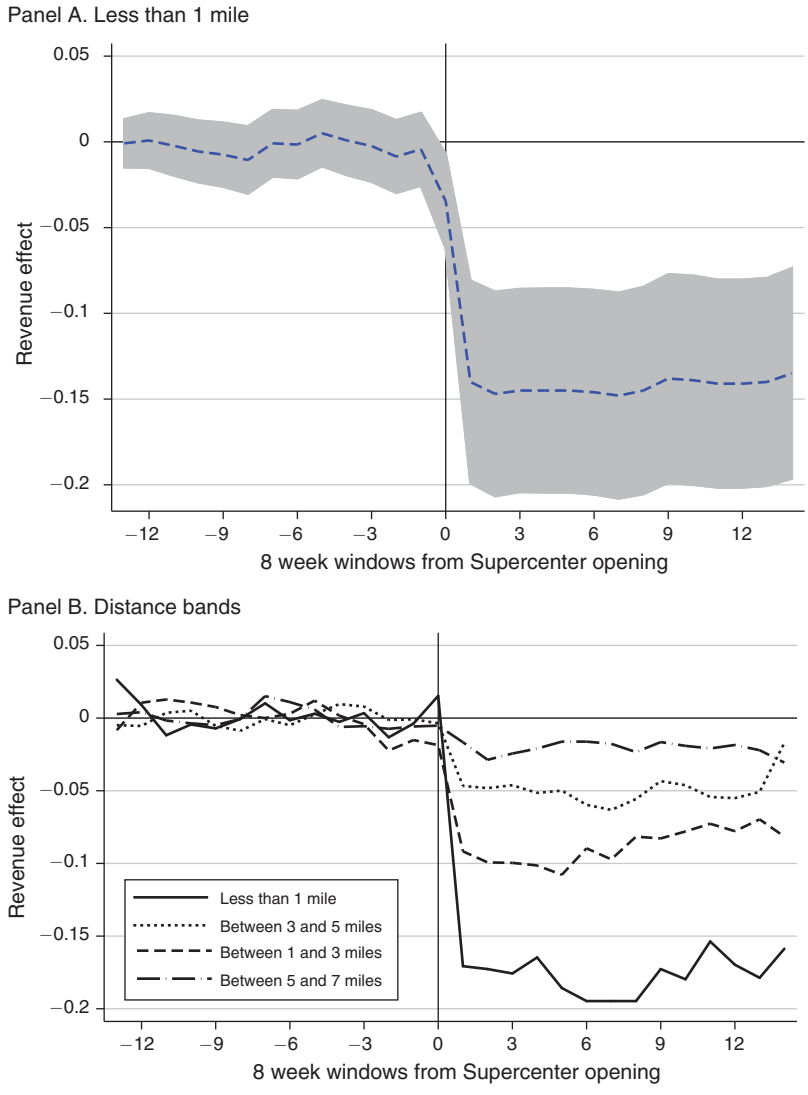


FIGURE 1. TRENDS IN THE REVENUE EFFECT OF SUPERCENTERS

exposure within 1 mile is associated with a massive 8 percent decrease in prices. Comparing incumbents in the same chain and market in column 2 attenuates the effect sizes some, with the results indicating about a 2–3 percent price reduction in response to a 1-mile exposure. These magnitudes align with those in the existing literature showing price effects on the order of 1–3 percent (Hausman and Leibtag 2007, Basker and Noel 2009). Notably, the estimated magnitudes also display a monotonic pattern with the exposure distance.

Isolating within-store changes in prices by conditioning on store fixed effects changes the results dramatically, however. Column 3 indicates that the price effects are attenuated such that they are statistically indistinguishable from zero

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Table 6—Competitive Price Effect of Walmart Supercenters

	(1)	(2)	(3)	(4)	(5)	(6)
#WM $<$ 1 mile	−0.0844	−0.0264	−0.00379	0.00388	0.00366	
	(0.0118)	(0.00534)	(0.00879)	(0.00706)	(0.00704)	
1 mile $\leq$ #WM $<$ 3 miles	−0.0582	−0.0155	−0.00318	0.00281	0.00204	
	(0.00671)	(0.00312)	(0.00604)	(0.00441)	(0.00440)	
3 miles $\leq$ #WM $<$ 5 miles	−0.0383	−0.00451	−0.00861	−0.00350	−0.00338	
	(0.00565)	(0.00242)	(0.00412)	(0.00317)	(0.00320)	
5 miles $\leq$ #WM $<$ 7 miles	−0.0151	−0.00414	−0.00911	0.00127	0.00138	
	(0.00451)	(0.00232)	(0.00356)	(0.00297)	(0.00291)	
7 miles $\leq$ #WM $<$ 9 miles	−0.00114	−0.00209	−0.00707	−0.00169	−0.00128	
	(0.00422)	(0.00208)	(0.00299)	(0.00235)	(0.00252)	
9 miles $\leq$ #WM $<$ 11 miles	−0.0111	−0.00206	−0.00425	0.000621	0.000218	
	(0.00424)	(0.00199)	(0.00288)	(0.00234)	(0.00238)	
#WM $<$ 5 miles						−0.000744
						(0.00241)
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Category fixed effects	Yes	Yes	No	No	No	No
Market and chain FE	No	Yes	No	No	No	No
Store-category FE	No	No	Yes	Yes	Yes	Yes
Category trends	No	No	Yes	Yes	Yes	Yes
Market trends	No	No	No	Yes	Yes	Yes
Treatment group trends	No	No	No	No	Yes	Yes

Notes: Table presents results of regressing incumbent supermarket log prices on Walmart Supercenter exposure. There are 11,253,196 store-category-week observations. Standard errors clustered by store in parentheses. Treatment groups are defined by total Supercenter exposures. Markets are IRI designated markets and category refers to product category. All trends are quadratic.

at the nearest distance bands. Columns 4 and 5, which incorporate various secular trends, find robust null effects. In our preferred specification (column 5), an estimated effect of Supercenter exposure on prices is not different from zero at any treatment distance, and further, the point estimates are very small in magnitude. Using the complete set of controls, column 6 pools all Supercenter exposures within 5 miles, the furthest distance at which we observe any revenue effects. The 95 percent confidence interval for this point estimate rules out even a modest 0.5 percent price decline due to Supercenter exposure, suggesting the null result is not a consequence of low statistical power.

Walmart's choice of where to locate rationalizes the sharp differences we find between results that control for store fixed effects and those that do not. Walmart chooses to locate next to low-price stores, so absent the inclusion of store fixed effects, treatment within the one-mile distance band serves as a proxy for being positioned as low-priced rather than identifying the causal effect of Supercenter entry on prices. To illustrate this, we recover the incumbent supermarkets' estimated fixed effects from column 5 of Table 6. These fixed effects represent each incumbent's prices absent the treatment effects of Supercenter exposure (and temporal factors). We decompose the fixed effects as a function of each incumbent's market, chain, local demographics, and nearest Supercenter exposure in Table 7. Column 1 shows

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Table 7—Predicted Prices Decomposition

	(1)	(2)	(3)	(4)	(5)
Closest WM $<$ 1 mile	−0.0390	−0.0469	−0.0444	−0.0428	−0.0266
	(0.0107)	(0.0113)	(0.0105)	(0.0104)	(0.00577)
1 mile $\leq$ closest WM $<$ 3 miles	−0.0333	−0.0356	−0.0343	−0.0347	−0.0176
	(0.00787)	(0.00805)	(0.00764)	(0.00748)	(0.00433)
3 miles $\leq$ closest WM $<$ 5 miles	−0.0209	−0.0261	−0.0316	−0.0309	−0.0102
	(0.00792)	(0.00808)	(0.00763)	(0.00756)	(0.00425)
5 miles $\leq$ closest WM $<$ 7 miles	−0.0187	−0.0271	−0.0290	−0.0300	−0.0195
	(0.00820)	(0.00813)	(0.00768)	(0.00755)	(0.00467)
7 miles $\leq$ closest WM $<$ 9 miles	−0.00277	−0.0124	−0.0177	−0.0186	−0.0129
	(0.0109)	(0.0111)	(0.0105)	(0.0103)	(0.00502)
9 miles $\leq$ closest WM $<$ 11 miles	−0.0133	−0.0218	−0.0235	−0.0225	−0.00273
	(0.0100)	(0.0105)	(0.00991)	(0.00961)	(0.00525)
Second income quintile			0.0343	0.0348	0.0268
			(0.00689)	(0.00690)	(0.00382)
Third income quintile			0.0531	0.0539	0.0414
			(0.00705)	(0.00705)	(0.00414)
Fourth income quintile			0.0906	0.0889	0.0620
			(0.00760)	(0.00754)	(0.00461)
Fifth income quintile			0.140	0.140	0.0983
			(0.00784)	(0.00777)	(0.00457)
Second density tercile			0.0126	0.0126	0.00998
			(0.00547)	(0.00534)	(0.00329)
Third density tercile			0.0226	0.0201	0.0139
			(0.00714)	(0.00696)	(0.00414)
Regional chain				0.0390	
				(0.00651)	
Subregional/local chain				0.0524	
				(0.0100)	
Non-chain supermarket				0.00340	
				(0.0109)	
Constant	−0.0117	0.00759	−0.0897	−0.100	−0.0300
	(0.0144)	(0.0155)	(0.0142)	(0.0159)	(0.0332)
Category fixed effects	Yes	Yes	Yes	Yes	Yes
Market fixed effects	Yes	Yes	Yes	Yes	Yes
Chain fixed effects	No	No	No	No	Yes
Observations	36,486	30,061	30,061	30,061	30,061

Notes: Table presents results of regressing incumbent supermarkets' price fixed effects on store and market characteristics. Price fixed effects are demeaned and correspond to regression in column 5 of Table 6. Robust standard errors are in parentheses. Markets are IRI-designated markets. National chain, first income quintile, and first density tercile stores are excluded group.

that stores exposed at nearer distances have lower predicted prices. Subsequent columns reveal this association is robust to comparing incumbent supermarkets in the same chain and market. The estimates in column 5 indicate that supermarkets exposed within 1 mile have nearly 3 percent lower predicted prices, while stores exposed between 1 and 3 miles have around 2 percent lower prices. Thus, rather than Supercenter exposure causing incumbents to lower prices, our findings show that Walmart locates near low-price competitors.

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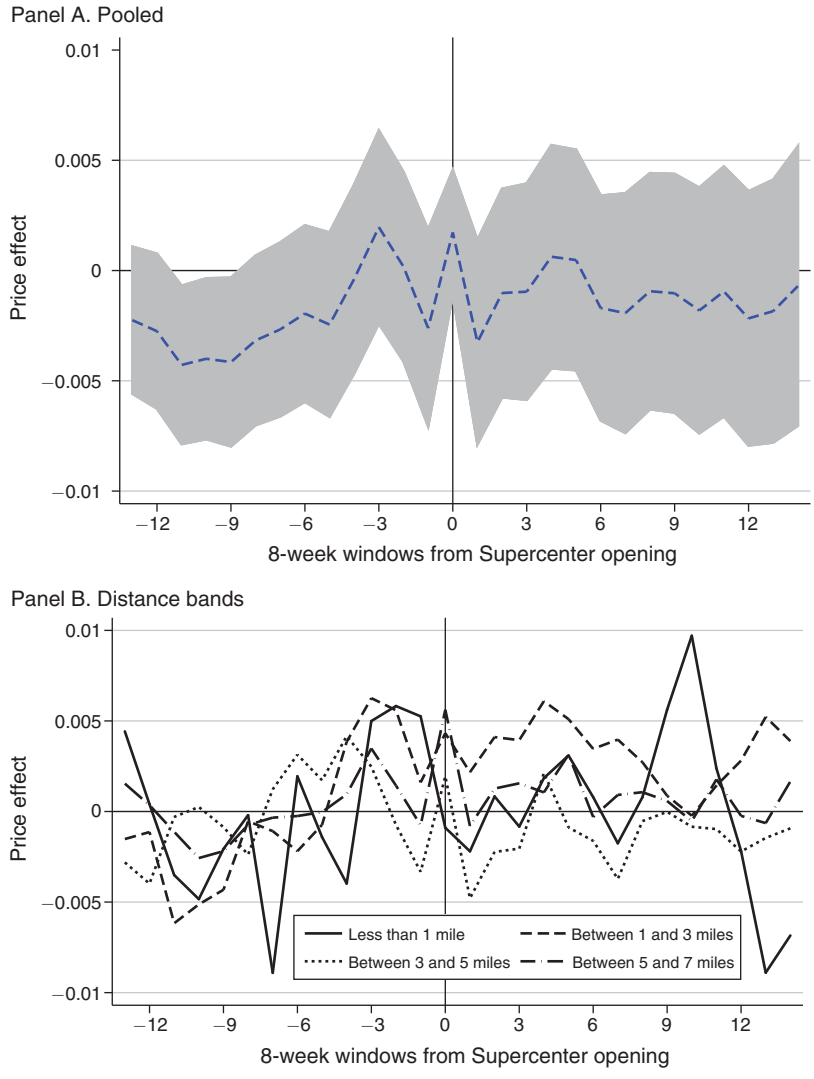


FIGURE 2. TRENDS IN THE PRICE EFFECT OF SUPERCENTERS

As with the revenue results, we check for differential preexposure trends and posttreatment effects on incumbent prices. The results from estimating our augmented "event study" model interacting treatment with time to exposure for exposures within 7-miles driving distance are displayed in online Appendix Table A2 and plotted in panel A of Figure 2.[31] Notably, there is neither evidence of pretreatment trends, validating causal inference, nor evidence of deviations from the null effect after entry. These conclusions are reinforced by the figure in panel B of Figure 2, which presents estimates that allow the trends to be distance band-specific.

[31] We do not include any controls for distance in these specifications.

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In addition, the standard errors of the treatment effects over time, depicted in panel A, are able to rule out even a modest 1 percent drop in prices postentry from the 95 percent confidence region. In other words, the finding of no posttreatment effect is reasonably precise and not simply due to lack of statistical power.

IV. Potential Confounds and Explanations

Our finding of no causal effect on incumbents' prices is surprising in light of the large revenue effect of Supercenter entry (as well as the prior results in the Walmart literature). Accordingly, in this section, we examine potential confounds and then evaluate several potential explanations for the nonresponse. We begin by examining whether changes in the composition of either products or consumers at incumbent supermarkets could either mask a meaningful response or lead to offsetting effects. We directly examine the effect of Supercenter entry on the assortment of products offered by incumbents and also estimate price effects at the UPC level. Across all specifications, we find consistent evidence of no price response. We then empirically evaluate several potential explanations for why firms would fail to adjust prices, including restrictive vertical contracts and chain- or zone-level pricing.

Though not discussed in the main text, we consider several additional confounds in the online Appendix. In online Appendix Table A3, we examine the effects of Supercenter entry on the likelihood that a product is offered at a discounted price at an incumbent supermarket. We also examine the robustness of our results to missing sales observations. In addition, we examine price effects corresponding to just the *first* Supercenter to which an incumbent is exposed in online Appendix Table A11; we find no evidence for price response to first-time Walmart Supercenter exposures.

A. *Product Assortment*

Product assortment at incumbent supermarkets may be affected by Supercenter entry. For example, incumbents may respond by offering smaller package sizes (that are relatively more expensive on a volume basis) or expanding product variety to differentiate from Walmart. In such a case, even if prices are cut, this change in assortment could offset price response and potentially be masked by our volume-weighted price index. So even if we observe no changes in our price index due to entry of a Walmart Supercenter, this may reflect a combination of the shift in product assortment that raises prices and a competitive price response that lowers them.

With this motivation, we investigate how Supercenter entry affects what products supermarkets sell in two ways. First, we examine how the total number of UPCs offered changes in response to Walmart entry. Here our dependent variable is the log of the number of unique UPCs sold across the fifteen product categories in each incumbent by quarter. We aggregate to quarters because some UPCs are slow-moving and may be particularly so following Supercenter entry. In this way, we limit the bias from supermarkets continuing to offer a product but having that product not sell. Around 2,700 unique UPCs are sold each quarter by the supermarkets in our

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Table 8—Competitive Assortment Effect of Walmart Supercenters

	log total UPCs		First quartile share		Fourth quartile share	
	(1)	(2)	(3)	(4)	(5)	(6)
#WM $<$ 1 mile	−0.00582		0.00179		−0.00185	
	(0.00857)		(0.00155)		(0.00250)	
1 mile $\leq$ #WM $<$ 3 miles	0.00881		0.000655		0.00261	
	(0.00691)		(0.00125)		(0.00186)	
3 miles $\leq$ #WM $<$ 5 miles	−0.00766		0.000954		−0.000943	
	(0.00484)		(0.00111)		(0.00111)	
5 miles $\leq$ #WM $<$ 7 miles	−0.00483		−0.000718		−0.00108	
	(0.00512)		(0.000916)		(0.00102)	
7 miles $\leq$ #WM $<$ 9 miles	−0.00441		0.00120		−0.000306	
	(0.00528)		(0.000955)		(0.000969)	
9 miles $\leq$ #WM $<$ 11 miles	−0.00535		−0.000228		−0.000271	
	(0.00461)		(0.000884)		(0.000852)	
#WM $<$ 5 miles		−0.000695		0.000839		0.000323
		(0.00405)		(0.000807)		(0.000972)

Notes: Table presents results of regressing incumbent supermarket log assortment and assortment shares on Walmart Supercenter exposure. Assortment shares measured as share of sales of unique UPCs that belong to the first (bottom) and fourth (top) quartiles of the category price distribution. There are 900,790 store-category-quarter observations. Standard errors clustered by store are in parentheses. All specifications include period fixed effects, store-category fixed effects, category trends, market trends, and treatment group trends. Treatment groups are defined by total Supercenter exposures. Markets are IRI-designated markets, and category refers to product category. All trends are quadratic.

sample.[32] Results of regressing the log of the number of unique UPCs are presented in the first set of columns in Table 8 using our full set of controls, both for different distant bands and estimating a common effect for entries within five miles. We do not find evidence that Supercenter entry has an effect on the number of unique UPCs offered.

Even though there is no change in the total number of UPCs on shelves, there may be changes in the composition of products offered. Our second test examines whether Walmart changes the share of UPCs offered that are lower or higher priced. For instance, incumbent supermarkets may respond to Supercenter entry by shifting or repositioning to higher-priced (higher-end) products to differentiate themselves from Walmart in product space. To examine the empirical relevance of this, we create price per volume quartiles for each product category and count the unique number of UPCs sold in the first (bottom) and fourth (top) quartiles in each quarter. We then regress the share of low- and high-price UPCs at the store-category-quarter level on Supercenter exposure and our controls. The results are presented in the last set of columns in Table 8. We find no evidence at any distance band (or aggregating across distance bands) that Supercenter entry affects the composition of products offered.

[32] Appendix Table A2 summarizes assortment at incumbent supermarkets.

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B. *UPC Aggregation*

Even if supermarkets do not change their product mix in response to Walmart entry, it is possible that the customers who continue to shop at these supermarkets may purchase more expensive bundles of products, which in turn may mute any negative price response in our analysis. So even if incumbents do not respond to Supercenter entry by changing what products are offered, the customers who remain may be more likely to purchase more expensive products or favor smaller package sizes than those who switch. In such a case, even if prices are cut, this effect could offset the price response and be masked by our volume-weighted price index.

To investigate the possibility, we implement our empirical strategy at the UPC level on four subsamples of the purchase data. In these results, the causal effect of Supercenter exposure is then identified by comparing prices for the same UPC in the same incumbent supermarket before and after Supercenter openings. This analysis thus tests whether changes in the composition of UPCs purchased postexposure possibly confound price changes, generating bias. For this item-level analysis, we consider the top 30 UPCs by total revenue sold during our sample for each of the carbonated beverage, margarine/butter, salty snack, and peanut butter categories (120 individual UPCs in total).

The results are displayed in Table 9 and reveal no robust evidence of price response to Supercenter exposure: conditioning on store fixed effects to isolate the changes that occur within treated stores as in our preferred specification presented in the odd columns, we find no evidence for competitive price response. The lone exception shown in Table 9 is a statistically significant price drop for peanut butter UPCs of about 1.6 percent for exposures within 1 mile. For all other distance bands, the effects are not different from zero, nor are they significant when all exposures within 5 miles are pooled, presented in the even columns. These UPC-level results therefore support the conclusion drawn from the aggregated price series of a null effect.

In online Appendix Table A3, we examine the effects of Supercenter entry on the likelihood that a product is offered at a discount at an incumbent and the robustness of our results to potentially missing sales observations. We find no effects on price promotion of UPCs and find similar price results when considering just the top 20 most frequently purchased UPCs (of the 120 examined in this subsection).

C. *Heterogeneity across Products*

A second potential confound is that exposure to Supercenter entry might generate a null causal effect by affecting price dispersion across products. This dual response could follow from a Supercenter changing the distribution of consumers who shop at the incumbent in addition to changing the competitive environment. For instance, a Supercenter opening may incentivize the store to price-compete for price-sensitive shoppers and to raise prices for brands demanded by quality-sensitive shoppers, leading to an offsetting effect that zeroes out across products. Along these lines, in a study of shopper "home scan" data, Jindal et al. (2018) find that low-priced "value" brands perform better at Walmart Supercenters, while premium brands perform

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TABLE 9—COMPETITIVE PRICE EFFECT OF WALMART SUPERCENTERS: TOP 30 UPCS

	Carbonated beverage		Margarine/butter		Peanut butter		Salty snack	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
#WM < 1 mile	0.00716		−0.00127		−0.0166		0.000653	
	(0.0103)		(0.00829)		(0.00815)		(0.00622)	
1 mile ≤ #WM < 3 miles	−0.00202		−0.00489		0.00509		−0.000370	
	(0.00679)		(0.00457)		(0.00443)		(0.00385)	
3 miles ≤ #WM < 5 miles	−0.000775		−0.00437		−0.00365		0.00156	
	(0.00499)		(0.00369)		(0.00262)		(0.00240)	
5 miles ≤ #WM < 7 miles	−0.00192		0.000127		0.00227		−0.000655	
	(0.00368)		(0.00302)		(0.00249)		(0.00242)	
7 miles ≤ #WM < 9 miles	−4.71e-05		−0.00122		0.00151		0.00175	
	(0.00344)		(0.00262)		(0.00220)		(0.00197)	
9 miles ≤ #WM < 11 miles	−0.00502		0.000252		0.00334		0.00283	
	(0.00342)		(0.00261)		(0.00233)		(0.00186)	
#WM < 7 miles		0.000338		−0.00420		−0.00302		0.000569
		(0.00360)		(0.00279)		(0.00228)		(0.00201)
Observations	15,626,919		13,767,512		11,032,410		7,723,576	

Notes: Table presents results of regressing UPC-level incumbent supermarket log prices on Walmart Supercenter exposure. For each product category, sample includes top 30 UPCs defined by total sales revenue in real terms. Standard errors clustered by store are in parentheses. All specifications include period fixed effects, store-UPC fixed effects, UPC trends, market trends, and treatment group trends. Treatment groups are defined by total Supercenter exposures. Markets are IRI-designated markets. All trends are quadratic.

better at supermarkets. Moreover, consumers who shop at both types of outlets are more price sensitive than those who are loyal to a single one. Similarly, a shifting consumer distribution may instead imply that the optimal incumbent response to a Supercenter opening is no response. The distributional change, termed the "price sensitivity" effect by Chen and Riordan (2008), implies the incumbent should *raise* prices, while the competitive or "market share" effect implies a price cut. The estimated null causal effect of entry on overall price would thus represent the net of these two offsetting effects when averaged across products.

We leverage across product variation to test these explanations of our findings. Intuitively, given Walmart's lower-end assortment offerings, the competition incentive is likely to dominate for low-quality products (purchased by price-sensitive consumers), while the market power effect should dominate on the high end. To test for this possibility, we therefore examine the effects of Supercenter entry across brands in product categories that exhibit strong vertical differentiation. As an example, frozen pizza is characterized by vertical differentiation across brands that is reflected in considerable price differences. Prices by brand are summarized in online Appendix Table A4, which shows that California Pizza Kitchen products are over three times as expensive on average as Totino's products in volume terms. In addition to frozen pizza, we examine effects across brands in the spaghetti sauce, coffee, and frozen dinner categories.[33]

[33] For each category, we extract the full universe of transactions records for UPCs that appear for at least 26 weeks and in 5 or more supermarkets (for nonprivate label items). For coffee, we remove "pod"-style packaged UPCs, introduced at the tail end of our estimation period, from the sample and adjust prices for serving size to account for "instant" varieties. We also restrict frozen dinner UPCs to entree-style products.

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Table 10—Competitive Effects of Walmart Supercenters by Brand: Frozen Pizza

	log revenue (1)	log price (2)
#WM × Amy's Kitchen	−0.0250	−0.00535
	(0.0350)	(0.00577)
#WM × CA Pizza Kitchen	−0.00131	0.00267
	(0.0368)	(0.00280)
#WM × Celeste	−0.0622	0.00332
	(0.0468)	(0.0133)
#WM × DiGiorno	−0.0643	−0.00100
	(0.0191)	(0.00207)
#WM × Freschetta	−0.0600	3.12e-05
	(0.0239)	(0.00335)
#WM × Jack's	−0.213	−0.00497
	(0.0741)	(0.00734)
#WM × Private Label	−0.0755	−0.000428
	(0.0411)	(0.00162)
#WM × Red Baron	−0.0725	−0.00272
	(0.0208)	(0.00339)
#WM × Stouffer's	−0.0572	−0.00352
	(0.0292)	(0.00320)
#WM × Tombstone	−0.0303	−0.00305
	(0.0316)	(0.00437)
#WM × Tony's	−0.0549	−0.00212
	(0.0308)	(0.00401)
#WM × Totino's	−0.0756	−0.00909
	(0.0279)	(0.00632)
#WM × other	−0.0843	0.000712
	(0.0380)	(0.00245)

Notes: Table presents results of regressing incumbent supermarket log revenues and log prices for frozen pizza on Walmart Supercenter exposure, allowing for heterogeneity by brand. WM = 1 for Supercenter exposures within 5 miles. There are 7,751,801 store-brand-week observations in column 1 and 75,467,137 store-UPC-week observations in column 2. Standard errors clustered by store in parentheses. Markets are IRI-designated markets. Column 1 includes period fixed effects, store-category fixed effects, category trends, market trends, and treatment group trends. Treatment groups are defined by total Supercenter exposures. All trends are quadratic. Column 2 includes store-UPC fixed effects and market-UPC-period fixed effects.

We first examine the effects of Supercenter exposure across brands in the frozen pizza category. The results obtained from applying our empirical model to revenues are presented in column 1 of Table 10. For major frozen pizza brands, we total revenue in each store and week and examine the effect of Supercenter exposure (pooling all entries within five miles). The estimates reveal that while some, generally lower-priced, brands experience sharp revenue drops in response to a Supercenter opening, other brands experience no significant change in revenue at all. For instance, revenues for Amy's Kitchen and California Pizza Kitchen, the highest-priced brands on average, do not appear to change postentry, whereas Totino's revenues decline by about 8 percent. Column 2 of Table 10 presents the corresponding results for the effects on frozen pizza prices. This model is applied at the UPC level to avoid biases arising

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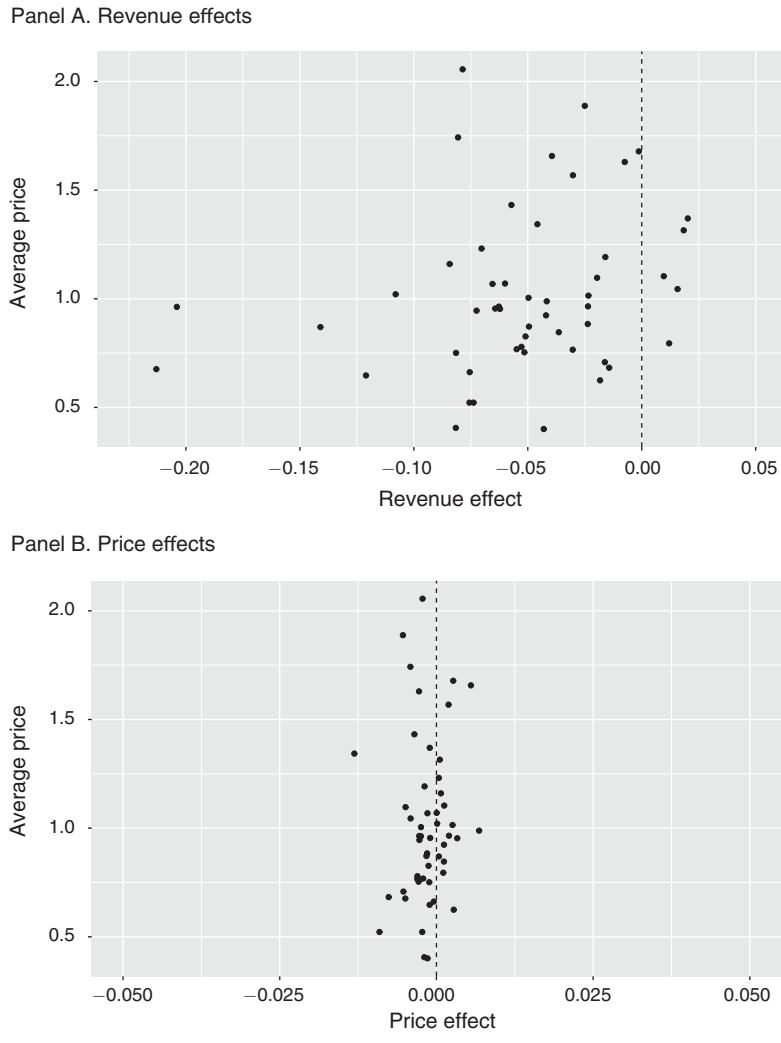


FIGURE 3. COMPETITIVE EFFECTS OF WALMART SUPERCENTERS BY BRAND

from aggregation.[34] For no brand is the point estimate statistically different from zero, indicating that despite the heterogeneity in revenue impacts across brands, there are no price effects.

To examine the robustness of these findings, we estimate revenue and price effects across brands for the spaghetti sauce, coffee, and frozen dinner product categories. The results are presented in the online Appendix and plotted for all four categories in Figure 3.[35] Panel A of Figure 3 plots the average price of each brand (relative to

[34] Note that the large sample size precludes the estimation of parametric trends. Estimation instead applies an iterative estimator with store-UPC fixed effects and market-UPC-period fixed effects.

[35] The online Appendix also summarizes the prices of UPCs for each brand.

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the category average) against the estimated revenue impact of Supercenter entry. The results reveal an upward-sloping pattern showing, as with the results for frozen pizza, that revenue effects are smaller for brands with higher average prices. In other words, lower-priced brands are generally hit hardest by Supercenter entry; fitting a line to the data points reveals a positive (and statistically significant) slope of 1.8. Panel B then displays average price plotted against price effects for each brand. In this case, no pattern is detectable, with the bulk of brands across all price tiers clustered near 0. These figures highlight that while demand shocks vary predictably across tiered brands in multiple categories, price reactions reveal no significant price response in general. These findings are inconsistent with "price sensitivity" and "market share" effects netting out.

D. *Heterogeneity across Private and Branded Labels*

We next assess the possibility that the null response is driven by the retailers not having full control of the pricing process as a potential explanation for our findings. In particular, it is conceivable that incumbent supermarkets are constrained in their price-setting ability by contractual (vertical) relationships with manufacturers. There is no clear consensus in the literature regarding whether the retailer or the manufacturer has the most power in setting prices. The previous literature has modeled this either as a noncooperative game (Villas-Boas 2007) or cooperative (Nash) bargaining problem (Draganska, Klapper, and Villas-Boas 2010). If the manufacturer dominates the relationship, the price response at an individual store may be muted.

To address this concern, we split our analysis between those branded products for which there is a potential channel conflict and private label, "store brand" products over which the retailer has direct control. In other words, it might be the case that although the prices of branded products are constrained by manufacturer relationships, incumbents can respond to a Supercenter opening by reducing the prices of private label products. The combined effect might appear as an overall null effect. To consider this possibility with our data, we construct category price series separately for all private label and branded UPCs using equation (1). For both branded and private labels, we then pool the logged category price series to estimate the price effect of Supercenter exposure. The results are displayed in columns 1 and 2 of Table 11. The specification we estimate incorporates store-category fixed effects and multiple secular trends while pooling all Supercenter exposures within 5 miles. The results reveal that at no distance band can the price effect be rejected as statistically different from zero at conventional thresholds for either branded or private products.

E. *Heterogeneity across Supermarket Type*

Next, we examine whether the nonresponse can be explained by retailer scale. Large chains may face incentives to attenuate their response to local entry to soften price competition as in a model of multimarket contact. Similarly, chain supermarkets may set relatively uniform retail prices across their stores within a given market

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Table 11—Competitive Price Effect of Walmart Supercenters by Branded, Private Labels, and Supermarket Type

	Branded (1)	Private label (2)	Type (3)
#WM	−0.000734	−0.00501	
	(0.00213)	(0.00357)	
#WM × national chain			−0.000696
			(0.00290)
#WM × regional chain			−0.00682
			(0.00610)
#WM × subregional/local chain			0.00599
			(0.00686)
#WM × non-chain			−0.00568
			(0.00701)
Observations	11,253,014	10,564,592	11,112,139

Notes: Table presents results of regressing incumbent supermarket log prices on Walmart Supercenter exposure for branded products, private label products, and allowing for heterogeneity by the type of the incumbent supermarket. WM = 1 for Supercenter exposures within five miles. Standard errors clustered by store are in parentheses. All specifications include period fixed effects, store-category fixed effects, category trends, market trends, and treatment group trends. Treatment groups are defined by total Supercenter exposures. Markets are IRI-designated markets. All trends are quadratic.

or region, either to economize on firm-level adjustment costs or to mitigate price comparisons within their chain.[36] This multimarket feature of price-setting would in turn limit the extent to which chains might optimally respond to Supercenter openings that only locally affect one or a few of their stores. Hence, we investigate whether price responses differ based on whether the incumbent supermarket belongs to a national chain, a regional chain, a subregional or local chain, or is an independent supermarket.

Using our pooled category price series, we interact incumbents' total Supercenter exposure with indicators for type to look for evidence of this form of heterogeneity. The results are displayed in column 3 of Table 11, which includes the full slate of controls and pools entries within five miles. The results indicate no statistically significant price reactions for any incumbent type. These results are thus consistent with our main finding of no price response by incumbent supermarkets to Supercenter openings.

[36] In an analysis of AC Nielsen data on store-level prices at US supermarkets, Nakamura (2008) finds that 65 percent of price variation is common to stores of a particular chain, while 17 percent is idiosyncratic to the store and product (the remaining 16 percent is common across stores of different chains selling the same product). This suggests that while some price changes occur at the level of the store, the majority are chain-wide (however, many of these are likely responses to chain-wide shocks). DellaVigna and Gentzkow (2019) find even higher levels of "uniform pricing" in the Kilt's Nielsen data, concluding that firms face chain-level decision-making costs. In a broad-ranging analysis of the same data, Hitsch, Hortacsu, and Lin (2017) also find that chain factors explain a larger fraction of price dispersion than market factors, which they conclude is partly driven by the fact that by segmenting the market, chains face relatively more homogeneous demand than the market as a whole. Consistent with our focus, Gagnon and López-Salido (2015) also find very limited *chain-level* price response to several types of demand shocks, including natural disasters, severe weather, and labor conflicts.

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F. *Cost-Plus Pricing*

The last explanation we consider is that supermarkets follow pricing strategies that do not adjust to demand shocks. That many firms adopt simple pricing rules has been well-known for quite some time, dating back at least as far as Cyert and March (1963). The strongest empirical evidence in this regard is provided by Eichenbaum, Jaimovich, and Rebelo (2011), who obtained data from a very large (over 1,000 stores) supermarket firm operating in the United States and Canada. Critically, they observe both shelf prices and wholesale costs, allowing them to connect price changes directly to input prices. Notably, they find that prices rarely change without an accompanying cost change, though prices are not always adjusted when costs move. Instead, they find that the firms adjust prices relatively infrequently with the apparent goal of maintaining relatively stable "target" markups over cost. In particular, the periodic adjustments are used to keep the actual markups within about plus or minus 10 percent of their long-run target.[37] Analyzing similar data from a different retailer, McShane et al. (2016) also find that when shelf prices are adjusted, it is typically in response to an associated cost shock, with the resulting price change used to maintain the prior gross margin. There is also a body of survey evidence documenting the ubiquity of cost-plus pricing, particularly amongst retail firms.[38]

In our setting, we find robust evidence for no price response—either downward, upward, or preemptively—by incumbent retailers to Supercenter entry. This finding is consistent with margin maintenance: as Supercenter entry does not change costs, it also does not trigger a change in the pricing rules used by incumbents. Further, our examination of price response across products provides two additional pieces of supporting evidence for this interpretation. First, we do not observe price reductions even for the lower-tier products whose sales are most impacted by Supercenter entry. In terms of the framework in Chen and Riordan (2008), these are the products for which competitive incentives to lower prices are likely to dominate. Although we cannot directly rule out the possibility that the incumbents Walmart locates nearest are unable to downward adjust prices, this seems unlikely. Table 7 shows that supermarkets located nearest to Supercenters have about 3 percent lower prices on average, but average margins are on the order of 30 percent (e.g., Stroebel and Vavra 2019). At the same time, the opposing "price sensitivity" effect suggests that increasing prices for the upper-tier products (many for which we do not see significant revenue impacts) is the optimal response, which we also do not find any evidence

[37] Analyzing the same dataset but focusing on stores close to the US-Canada border, Gopinath et al. (2011) find that almost all of the variation in relative retail prices, in response to exchange rate shocks, is explained by variation in relative costs and not by variation in relative markups.

[38] An example is Watson, Wood, and Fernie (2015), who interview grocery pricing managers in the United States and the United Kingdom and find that minimum margin targets are the predominant practice in both countries. They find that competitive-based pricing strategies (i.e., reacting to rival prices) are typically viewed by practitioners as being infeasible due to poor access to data for competing firms and the inherent challenge of tracking a vast array of prices across a large number of competitors with a limited staff and budget. In addition, Phillips (2005) summarizes a host of previous surveys on pricing practices, noting that cost-plus pricing remains a popular pricing strategy that is used by up to 70 percent of firms in some geographies. Additional survey evidence is provided in Blinder et al. (1998) and Noble and Gruca (1999).

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for. Although this evidence is not itself conclusive (cross price elasticities must factor in, and upward adjustment could be constrained by reputation considerations), cost-plus pricing provides a consistent rationalization. While our data, which are lacking information on costs, do not allow us to test for cost-plus pricing, the fact that we find that incumbent firm size does not change our null result suggests that it is not simply a fixed (chain-level) adjustment cost but rather a broad-based pattern of behavior.

Our results suggest that such "cost-plus" thinking may influence how firms respond to even large-scale changes in their competitive environment. In the retail setting, simple rules of thumb might be an optimal response to an intractable optimization problem that involves monitoring the prices of thousands of products at numerous competitors while managing one's own complex supply chain. The intractability of this problem is further augmented by the practical challenges in many settings of collecting and analyzing vast amounts of competitor and consumer data. While advances in information technology and data science have substantially lessened the burden of solving such problems, our findings suggest that even the most sophisticated retailers are still unable to quickly adapt.

V. Conclusion

We combine data on weekly grocery transactions with the exact locations and opening dates of Walmart Supercenters to examine competitive price reactions by incumbent supermarkets. The unique temporal and spatial variation in our data allows us to separate the causal effect of entry from Walmart's decision of where to locate. While we find large but localized impacts on revenues, we find no evidence that incumbent supermarkets adjust prices in response to Supercenter entry. This result is robust across many dimensions, including a lack of price response preemptively, for individual products, and across brands within a category. Our findings stand in sharp contrast to earlier studies that suggest sizable price reductions that imply considerable welfare gains to consumers via competitive mechanisms (Hausman and Leibtag 2007, Basker and Noel 2009).

We argue that this nonresponse is unlikely to be explained by menu or monitoring costs, vertical contracts with manufacturers, or by multimarket pricing but is most consistent with widespread use of simple, "rule of thumb" cost-plus pricing strategies, a form of managerial inattention. Since entry by Walmart does not impact costs, it does not trigger a price reaction from incumbent supermarkets. This conclusion raises the question of why such simple policies persist in competitive environments and how much profit is left on the table by not employing more sophisticated strategies. Assessing the implications of these practices on both profitability and consumer surplus (via a demand model or using data that provide information on costs or survival) and how more recent entrants, in particular competition with online grocery retailers, may influence strategies are promising areas for future research, as is understanding of how such rules may be initially formed and updated.

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Appendix

Table A1—Incumbent Supermarket Characteristics

	N	Mean	SD	p10	p25	Median	p75	p90
Initial exposures to WM	2,450	1.31	1.36	0	0	1	2	3
Total exposures to WM	2,450	1.81	1.75	0	1	1	3	4
National chain	2,450	0.60	0.49	0	0	1	1	1
Regional chain	2,450	0.16	0.36	0	0	0	0	1
Subregional/local chain	2,450	0.12	0.32	0	0	0	0	1
Non-chain supermarket	2,450	0.06	0.24	0	0	0	0	0
Household median income	2,012	58,499	20,308	36,570	44,505	54,577	69,066	85,154
Population density	2,014	3,473	3,993	685	1,381	2,530	4,265	6,669

Notes: Table summarizes incumbent supermarket characteristics. Each observation represents an incumbent. Household median income and population density represent 2007 estimates for a two-mile radii around each store.

Table A2—Summaries of Incumbent Supermarket Product Assortment

	Mean	SD	p10	p25	Median	p75	p90
Total UPCs	2,686.21	642.21	1,907	2,224	2,655	3,113	3,561
First quartile UPCs	710.29	167.59	514	603	702	811	917
Fourth quartile UPCs	1,071.00	408.76	590	759	1,006	1,336	1,677

Notes: Table summarizes assortment, measured by sales of unique UPCs, at the incumbent supermarket-quarter observation level. First and fourth quartile counts represent unique UPCs in bottom and top quartile of category price distribution. There are 60,152 store-quarter observations.

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